BIO-DATA / CURRICULUM VITAE

DR. ANIKA SHARMA

Scientist E — Atmospheric Instrumentation & Applied AI

anika.sharma@example.org | +91 00000 12345

1. PERSONAL AND OFFICIAL PARTICULARS

Name	Anika Sharma
Present designation	Scientist E
Present organisation	Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory
Present pay level	Level 12 (7th CPC)
Email	anika.sharma@example.org
Telephone	+91 00000 12345

2. EDUCATIONAL QUALIFICATIONS

Qualification	Discipline / Subject	Institution / University	Year	Division / Grade
B.E.	Electronics and Communication Engineering	Northfield Institute of Science and Technology, Northfield University	2004	First Class with Distinction
M.Tech.	Avionics Thesis: Fault-tolerant data acquisition for airborne test instrumentation	Coastal Institute of Aeronautical Engineering	2010	CGPA 8.6/10
Ph.D.	Electrical and Electronics Engineering Thesis: Lightweight deep networks for real-time enhancement of aerial imagery	Northfield Institute of Science and Technology	2021 – ongoing	Ongoing

3. DETAILS OF EMPLOYMENT (IN CHRONOLOGICAL ORDER)

Office / Organisation	Post held	From	To	Pay level	Nature of duties
Test Systems Group, Meridian Electronics Limited	Graduate Engineer	08/2002	05/2004		Built automated test equipment for avionics line-replaceable units.
Avionics Integration Group, Halcyon Avionics Design Bureau	Senior Design Engineer	06/2004	06/2016		Design, integration and environmental qualification of avionics line-replaceable units; development of automated test rigs; support to flight trials of UAV payloads.

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Office / Organisation	Post held	From	To	Pay level	Nature of duties
Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory	Scientist E	07/2016	Till date	Level 12	Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; management of the division workshop; leadership of IoT monitoring and AI/ML development; technical evaluation and procurement of instruments under GFR; guidance of project staff and interns. Additional charge: Workshop In-charge, Upper Air Division Workshop (04/2019 – Present) Nodal officer: Nodal Officer, IoT and AI/ML Initiatives (01/2021 – Present)

4. PUBLICATIONS

[1] A. Sharma and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] A. Sharma, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] A. Sharma and R. Iyer, “Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs,” Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] A. Sharma, P. Rao and S. Gill, “An IoT telemetry framework for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] A. Sharma and S. Verma, “Edge telemetry dashboards for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] A. Sharma, P. Menon and R. Iyer, “A low-cost GNSS receiver chain for radiosonde tracking,” Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

5. MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] A. Sharma, M. Khan and R. Iyer, “Detection-aware quality assessment of super-resolved drone images,” under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] A. Sharma, “Simulation study of balloon-released gliding sondes,” in preparation. *[In preparation]*

6. MAJOR PROJECTS AND TECHNICAL CONTRIBUTIONS

Edge super-resolution pipeline for UAV imagery *(Ongoing)* 09/2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol 01/2018 – 03/2019

Coordinator

Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

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IoT monitoring of laboratory infrastructure 02/2020 – 06/2022

Project lead

Instrument rooms lacked continuous environmental and power-quality monitoring.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

7. TRAINING PROGRAMMES

Programme / course	Institute / sponsor	Period	Duration
Mid-Career Training Programme for Scientists	Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences)	02/2020	10 days
Public Procurement and GFR 2017	Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory)	11/2019	5 days

8. PROFESSIONAL COURSES

Programme / course	Institute / sponsor	Period	Duration
Introduction to Machine Learning (12-week course)	Example Online University Consortium	10/2018	48 hours

9. CERTIFICATIONS & LICENCES

Deep Learning for Edge Devices 03/2022

Example Academy of Applied Computing

Used the model-compression techniques to deploy the drone super-resolution model on Jetson modules.

Duration: 40 hours
Credential: EAAC-DL-0042
Valid until: 03/2029
Mode: Online
Skills: Deep learning, Edge AI, TensorRT

Academic English proficiency test *(Expired)* 05/2016

Example Language Testing Board
Result: Overall band 8.0
Expired: 05/2018

10. WORKSHOPS, SEMINARS & CONFERENCES ATTENDED

Programme / course	Institute / sponsor	Period	Duration
Workshop on Efficient Deep Learning for Embedded Vision	Example Conference on Instrumentation (ECI 2023)	12/2023	2 days
National Seminar on Upper-Air Observing Systems	Example Society of Atmospheric Instrumentation	08/2021	1 day

11. AWARDS & HONOURS

Best Paper Award 2023

Example Conference on Instrumentation (ECI 2023)

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12. ADDITIONAL RESPONSIBILITIES AND COMMITTEES

Supervisor, M.Tech. interns in applied machine learning 2022 – Present
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement 04/2023 – 02/2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

13. DECLARATION

I hereby declare that the particulars furnished above are true, complete and correct to the best of my knowledge and belief.

Place: New Delhi
Date: (Signature)
Anika Sharma

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